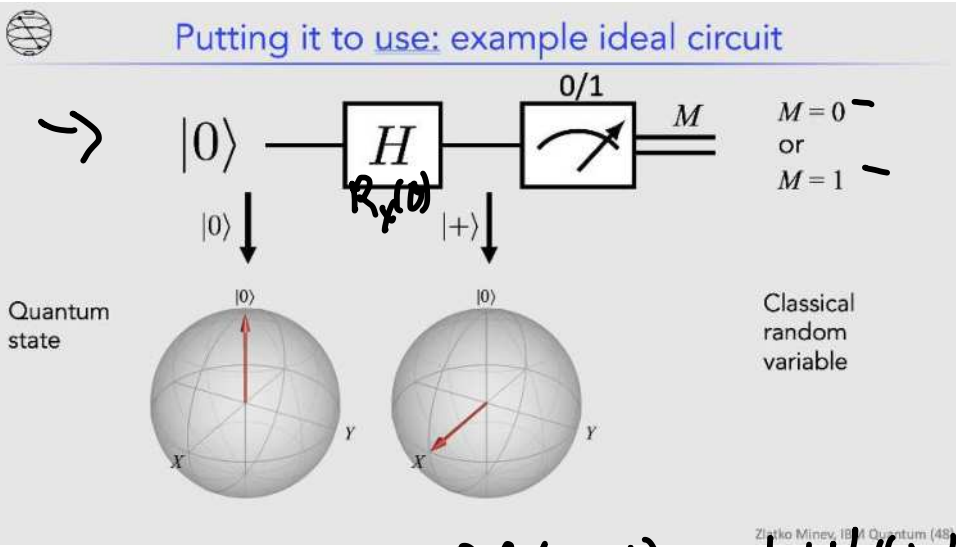
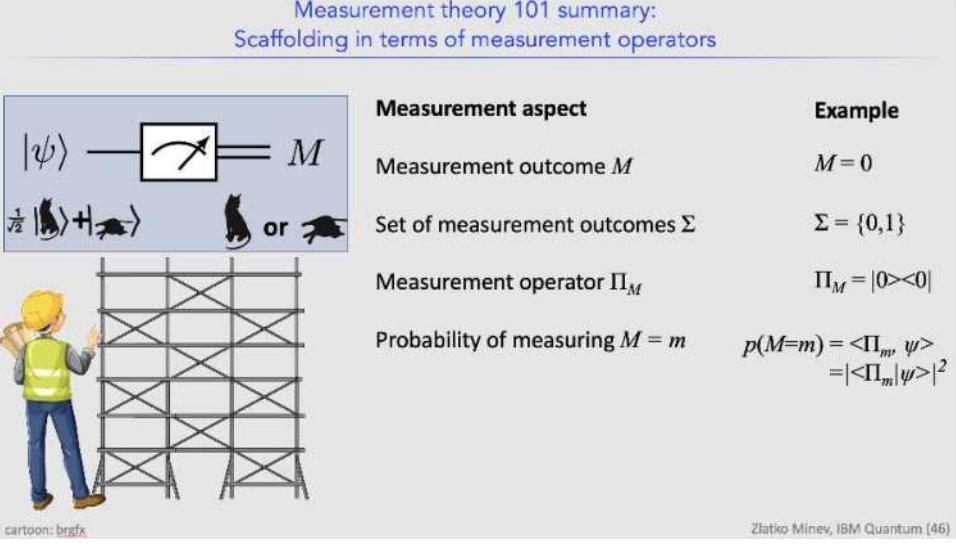
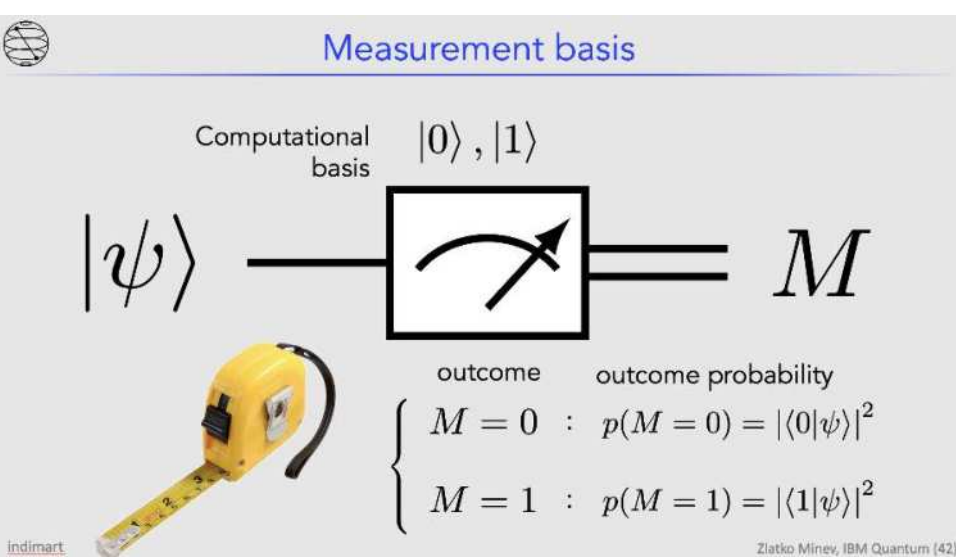


Introduction to quantum noise  
Measurement theory & projection noise

Qiskit Global Summer School on Quantum Simulations

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$\hat{\Pi}_0 = |0\rangle\langle 0|$   
 $\hat{\Pi}_1 = |1\rangle\langle 1|$   
 $|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$   
 $|\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^+$   
 $= \begin{pmatrix} 1 & 0 \end{pmatrix}$

$p(M=1) = |\langle 1|\psi\rangle|^2$   
 $p(M=0) = |\langle 0|\psi\rangle|^2$   
 $= \langle \psi | \hat{\Pi}_1 | \psi \rangle = \hat{\Pi}_0 | \psi \rangle$   
 $= \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$   
 $= \frac{1}{2} \cdot 1 = \frac{1}{2}$

$p(M=1) + p(M=0) = 1 \Rightarrow p(M=1) = 1 - p(M=0)$   
 $M: \begin{cases} 1 & p=1/2 \\ 0 & p=1/2 \end{cases}$   
Prob  $\begin{matrix} 1/2 & 1/2 \\ \uparrow & \uparrow \\ M=0 & M=1 \end{matrix}$

Statistics: Mean  
Random variable  $M$  drawn from distribution  
 $\mathbb{E}[M] = \sum_{m \in \Sigma} m p(M=m)$   
classical expectation value  
 $= 0 \cdot p(M=0) + 1 \cdot p(M=1) = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$

more general case

Prob to find  $M=1$ :  $p(M=1) = p$   $p \in [0, 1]$

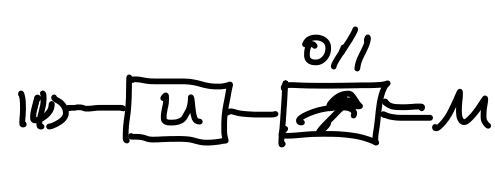
every possible qubit case

Bernoulli Distribution

$p(M) \sim B(p)$

$\uparrow$  classical random variable  
 $\uparrow$  probability

Prob  $\begin{matrix} 1-p & p \\ \uparrow & \uparrow \\ M=0 & M=1 \end{matrix}$



$\mathbb{E}[M] = \sum_{m \in \Sigma} m p(M=m)$  classical  
 $= 0 p(M=0) + 1 p(M=1)$   
 $= 0(1-p) + 1 \cdot p$   
 $= p$

$\mathbb{E}[M] = \sum_{m \in \Sigma} m p(M=m)$  quantum  
 $= \sum_{m \in \Sigma} m \langle \hat{\Pi}_m \rangle$   
 $= \sum_{m \in \Sigma} m \langle \psi | \hat{\Pi}_m | \psi \rangle$   
 $= \langle \psi | \sum_{m \in \Sigma} m \hat{\Pi}_m | \psi \rangle$   
 $= \langle \psi | \hat{M} | \psi \rangle$   
 $= \langle \hat{M} \rangle$  observable

$\langle \hat{\Pi}_m \rangle = \langle \psi | \hat{\Pi}_m | \psi \rangle = \langle \psi | m \rangle \langle m | \psi \rangle$   
linear  
 $\langle a \hat{B} \rangle = a \langle \hat{B} \rangle$   
 $\langle \hat{A} + \hat{B} \rangle = \langle \hat{A} \rangle + \langle \hat{B} \rangle$   
define  
 $\hat{M} := \sum_{m \in \Sigma} m \hat{\Pi}_m$  defn

Different observables  
For example  $\hat{M} = 0|0\rangle\langle 0| + 1|1\rangle\langle 1| = \frac{1}{2}(\hat{\sigma}_z + \hat{I}) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$   
 $\hat{M} = \frac{1}{2}(|0\rangle\langle 0| - |1\rangle\langle 1|) = \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$   
for  $\hat{M} = \hat{Z}$   $\Sigma = \{-1, +1\}$  Pauli observables



$\hat{M} = \hat{X}$

$\mathcal{S} = \{+, -\}$

$\hat{\Pi}_+ = |+\rangle\langle +|$

$\hat{\Pi}_- = |-\rangle\langle -|$

$\hat{X} = \sum_{m \in \mathcal{S}} m |m\rangle\langle m|$

$= +|+\rangle\langle +| - |-\rangle\langle -|$

$\hat{X}|+\rangle = +|+\rangle$

$\hat{X}|-\rangle = -|-\rangle$

$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Statistic: Variance

$V[M] = E[M^2] - E[M]^2$

$= \langle \hat{M}^2 \rangle - \langle \hat{M} \rangle^2$

$\langle \hat{M} \rangle = \langle \psi | \hat{M} | \psi \rangle$

$\langle \hat{M}^2 \rangle = \langle \psi | \hat{M} \cdot \hat{M} | \psi \rangle$

Return to  $\mathcal{S} = \{0, 1\}$

$\hat{\Pi}_0 = |0\rangle\langle 0|$

$\hat{\Pi}_1 = |1\rangle\langle 1|$

$p(M=1) = p$

$E[M^2] = \sum_{m \in \mathcal{S}} m^2 p(M=m)$

$= \sum m^2 \langle |m\rangle\langle m| \rangle$

$= \langle \sum m^2 |m\rangle\langle m| \rangle$

$= \langle \hat{M}^2 \rangle$

$= \langle |1\rangle\langle 1| \rangle$

$= \langle \hat{\Pi}_1 \rangle$

$= p = E[M]$

$V[M] = p - p^2$

$= p(1-p)$

$= \sigma_M^2$

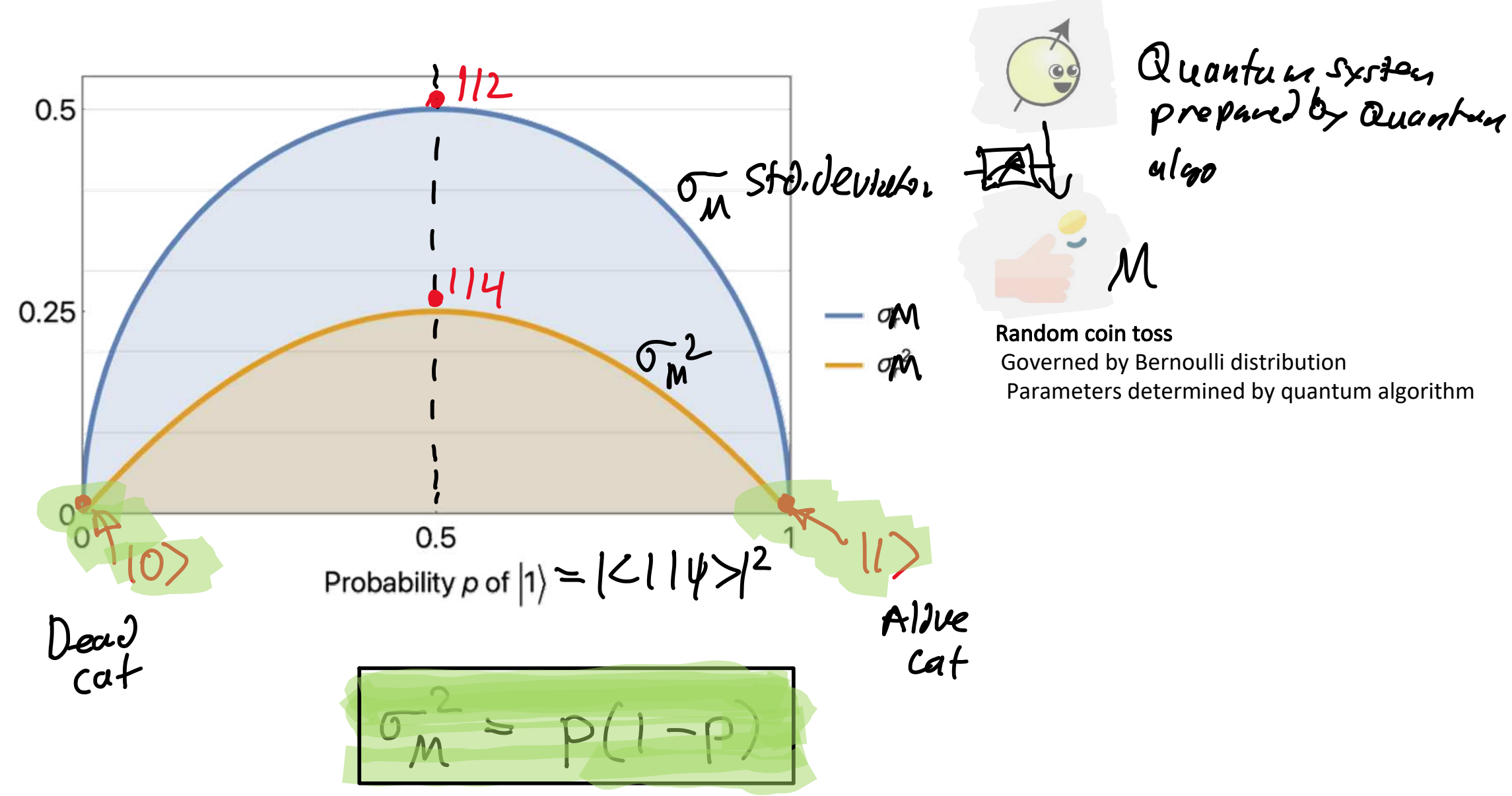
$p(M=m) = \langle |m\rangle\langle m| \rangle$

$\hat{M} = 0|0\rangle\langle 0| + 1|1\rangle\langle 1|$

$\hat{M}^2 = 1|1\rangle\langle 1| = M$

0 iff  $p=0$  or  $p=1$

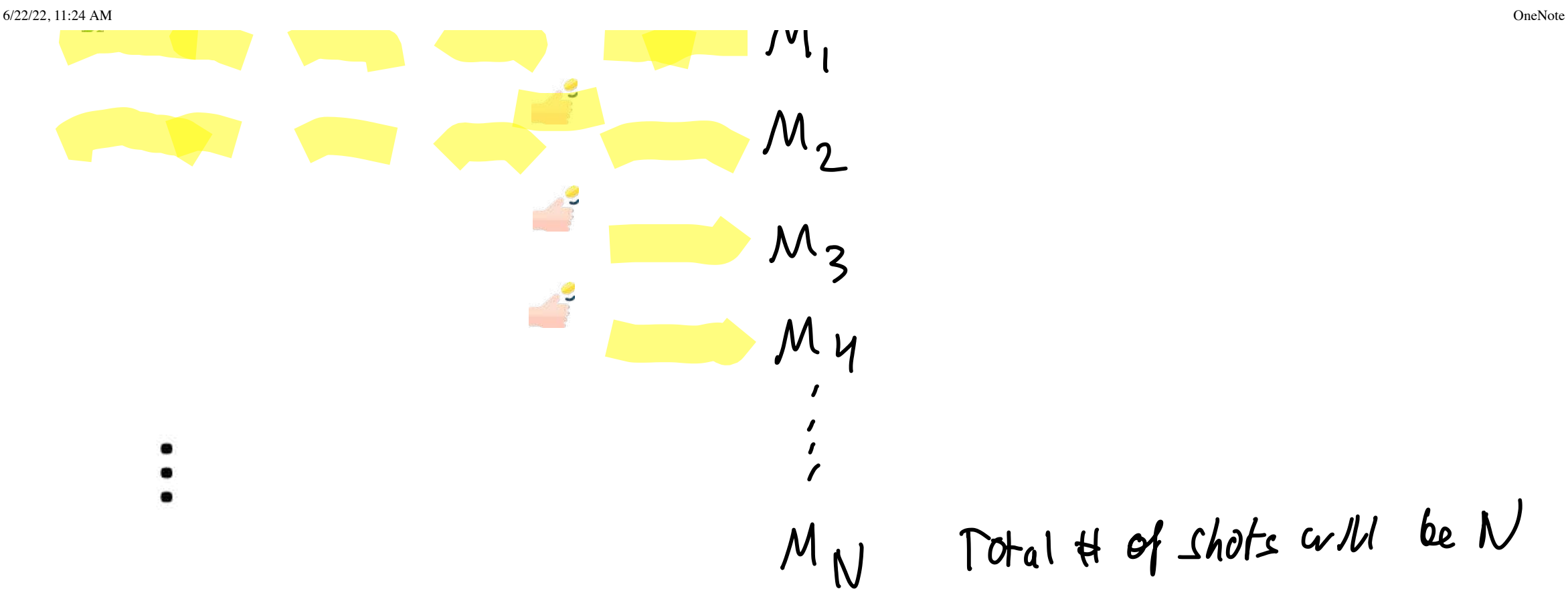
Variance of the random classical variable vs. probability to obtain 1



Projection noise and sampling error

Let's turn to the example of finite number of shots we execute for our experiment. Perform N experiment, each giving us a single shot result 0 or 1





For 3 samples, there are  $2^3$  possible outcome sequences ( $N=3$ )

$m_1 \rightarrow \begin{matrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{matrix}$ 
 $m_2 \rightarrow \begin{matrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{matrix}$ 
 $m_3 \rightarrow \begin{matrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{matrix}$

Sample mean?

$$S := \frac{1}{N} \sum_{n=1}^N M_n$$

$\nwarrow$  random variable  
 $\nwarrow$  index of the shot #

$\nwarrow$  # shot

indep. identically distributed (i.i.d.)

$M_n \sim M$

$2^N$  possible combinations  $\rightarrow$  examp  $S = \frac{1}{3}(1+0+1) = 2/3$

$$\begin{aligned}
 \mathbb{E}[S] &= \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N M_n\right] \\
 &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[M_n] \quad \mathbb{E}[M_n] = \mathbb{E}[M_1] = \mathbb{E}[M] \\
 &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[M] \\
 &= \mathbb{E}[M]
 \end{aligned}$$

$\mathbb{E}[S] = \mathbb{E}[M] = \langle \hat{M} \rangle = p$  for qubit  $\hat{M} = |1\rangle\langle 1|$

$$V[S] = \frac{V[M]}{N} = \frac{p(1-p)}{N}$$

$\sigma_S = \sqrt{\frac{p(1-p)}{N}}$   $\nwarrow$  #shots

error bar (std dev) on sample mean estimator for our expectation value.